

Supplementary Methods

Dispersion, not displacement: loss of alveolar epithelial state coherence in lethal COVID-19

1 Donor-level inference

1.1 Donor summary statistics

For each of the 27 donors, we computed the following cell-level summary statistics by aggregating across all alveolar epithelial cells from that donor:

- **Median pseudotime:** median diffusion pseudotime across all cells from the donor.
- **IQR pseudotime:** interquartile range of pseudotime.
- **Within-donor dispersion:** variance of Euclidean distances from each cell to the donor's centroid in 50-dimensional PCA space.
- **Median distance to donor centroid:** median Euclidean distance from cells to the donor's PCA centroid.
- **Fraction transitional:** fraction of cells annotated as ECM-high epithelial.
- **Program scores:** median score for each of 8 gene programs.

1.2 Donor-level statistical tests

All primary comparisons use the donor (not the cell) as the inferential unit. For each metric, we performed a two-sided Mann–Whitney U test comparing COVID-19 donors ($n = 20$) against healthy donors ($n = 7$). Effect sizes are reported as rank-biserial correlations: $r = 1 - 2U/(n_1n_2)$.

1.3 Bootstrap confidence intervals

We computed 10,000-replicate bootstrap confidence intervals for the difference in donor-level medians between conditions. At each replicate, we resampled donors (not cells) with replacement within each condition group and computed the difference in group medians. The 95% CI was taken as the 2.5th and 97.5th percentiles of the bootstrap distribution.

Results:

- Pseudotime: observed difference +0.041, 95% CI $[-0.013, 0.086]$
- Within-donor dispersion: observed difference +6.84, 95% CI $[-30.7, 16.5]$
- Fraction transitional: observed difference +0.009, 95% CI $[-0.053, 0.020]$

The bootstrap CI for pseudotime marginally includes zero, reflecting the low power of a 7-versus-20 comparison. The cell-level permutation test ($p = 0.001$) and cross-cohort replication ($p = 0.002$ at donor level) provide independent corroboration.

1.4 OLS donor-level model

We fit an ordinary least-squares regression: $\text{median_pseudotime} \sim \text{is_covid}$, treating each donor as one observation ($n = 27$). A random intercept for donor is unnecessary at this level because each donor contributes exactly one observation.

Results: COVID-19 coefficient = +0.043 (95% CI $[0.010, 0.076]$; $p = 0.013$; $R^2 = 0.22$).

1.5 Donor-level dispersion comparison

For each donor, we computed the median Euclidean distance from the donor’s cells to the donor’s centroid in PCA space. We then compared per-donor median dispersions between conditions using a one-sided Mann–Whitney U test (alternative: COVID > Healthy) and Levene’s test on the per-donor dispersion values.

Results: COVID-19 donors had median per-donor dispersion = 7.77; healthy donors had = 10.58 (Mann–Whitney $p = 0.999$; Levene $p = 0.071$). This indicates that the cell-level dispersion finding (Levene $p < 10^{-30}$) arises from between-donor heterogeneity within the COVID-19 condition, not from uniform within-donor inflation.

2 Portable state scores

Six continuous state scores were computed for each cell:

1. **AT2 identity:** mean expression of AT2 markers (SFTPC, SFTPA1, SFTPA2, ABCA3, LAMP3, SLC34A2, NAPS A, PGC, LPCAT1).
2. **AT1 identity:** mean expression of AT1 markers (AGER, PDPN, HOPX, CLIC5, CAV1, EMP2, AKAP5, RTKN2).
3. **Transitional:** mean expression of transitional markers (KRT8, CLDN4, SFN, KRT18, LGALS3, HOPX, AGER, PDPN) from the AT2-to-AT1 differentiation program.
4. **Injury composite:** mean of per-cell scores for the six non-homeostatic programs (interferon, NF- κ B, oxidative stress, apoptosis, senescence, AT2-to-AT1 differentiation).

5. **Coherence loss:** Euclidean distance from each cell to the centroid of healthy cells in PCA space.
6. **Repair failure:** transitional score $-0.5 \times (\text{AT2 identity} + \text{AT1 identity})$.

2.1 Benchmark results (primary atlas)

Score	$\rho(\text{pseudotime})$	MW p (condition)	COVID median	Healthy median	AUC (transitional)
AT2 identity	-0.693	1.7×10^{-25}	0.650	0.605	0.323
AT1 identity	+0.687	5.7×10^{-5}	0.394	0.348	0.375
Transitional	+0.282	1.3×10^{-20}	0.285	0.278	0.441
Injury composite	+0.276	3.8×10^{-70}	-0.007	-0.024	0.630
Coherence loss	+0.412	1.1×10^{-72}	10.68	11.56	0.972
Repair failure	+0.265	1.3×10^{-55}	-0.342	-0.407	0.730

Table 1: State score benchmarks on primary atlas. ρ : Spearman correlation with pseudotime. MW: Mann-Whitney U test between conditions. AUC: area under ROC curve for detecting annotated transitional cells.

The coherence-loss score achieves $\text{AUC} = 0.972$ for transitional-cell detection, confirming that geometry-based scores capture the transitional phenotype without requiring marker thresholds.

2.2 Replication cohort state scores

All six state scores were computed on the independent replication cohort (89,736 cells, 618 donors) using available genes from the 84-gene panel. PCA was recomputed on the replication data. All scores showed significant condition separation ($p < 10^{-300}$), confirming that the continuous, manifold-based scoring approach transfers across cohorts where marker-threshold DATP calling fails.

3 Program-geometry linkage

For each of 8 gene programs, we computed:

1. Spearman ρ with Euclidean distance to healthy centroid in PCA space.
2. Spearman ρ with diffusion pseudotime.

AT2 identity was most strongly anti-correlated with both distance ($\rho = -0.18$) and pseudotime ($\rho = -0.70$). AT2-to-AT1 differentiation was the strongest positive correlate of pseudotime ($\rho = 0.31$) and distance ($\rho = 0.17$).

3.1 Fan-out contribution

We decomposed the fan-out by regressing distance-to-healthy-centroid on all 8 program scores (linear regression, $R^2 = 0.115$). Top contributors: senescence (coefficient = 6.83), oxidative stress (3.83), NF- κ B (1.95). Negative contributors: AT2 identity (-1.13), AT1 identity (-1.02). This indicates that the dispersion pattern is driven by divergent activation of stress/senescence programs across cells.

3.2 Repair-stall analysis

We computed the ratio of AT2-to-AT1 differentiation score to apoptosis score in 10 equal-width pseudotime bins, stratified by condition. In healthy controls, the differentiation/apoptosis ratio was consistently higher than in COVID-19 at late pseudotime, consistent with a model where COVID-19 cells initiate repair but are diverted toward apoptotic rather than differentiation endpoints.

4 Software

All analyses were performed in Python 3.12.3 using: Scanpy 1.12.1, AnnData 0.12.10, NumPy 2.4.3, SciPy 1.17.1, pandas 2.3.3, statsmodels 0.14.6, scikit-learn 1.8.0. Random seeds fixed to 42 throughout.

5 Supplementary Figures

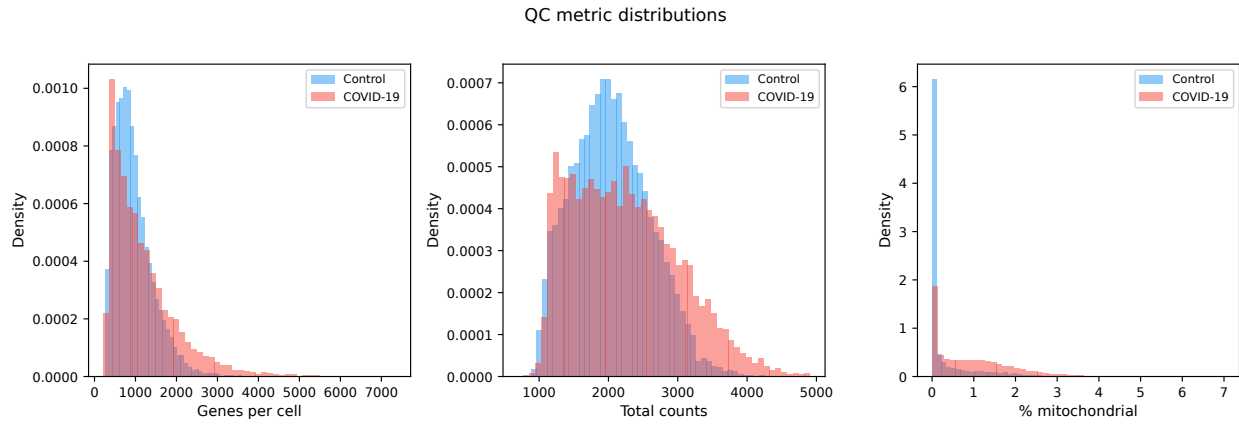


Figure 1: **Figure S1: Quality control.** Per-donor cell counts, gene counts, and mitochondrial fraction distributions for the 22,128 alveolar epithelial cells across 27 donors.

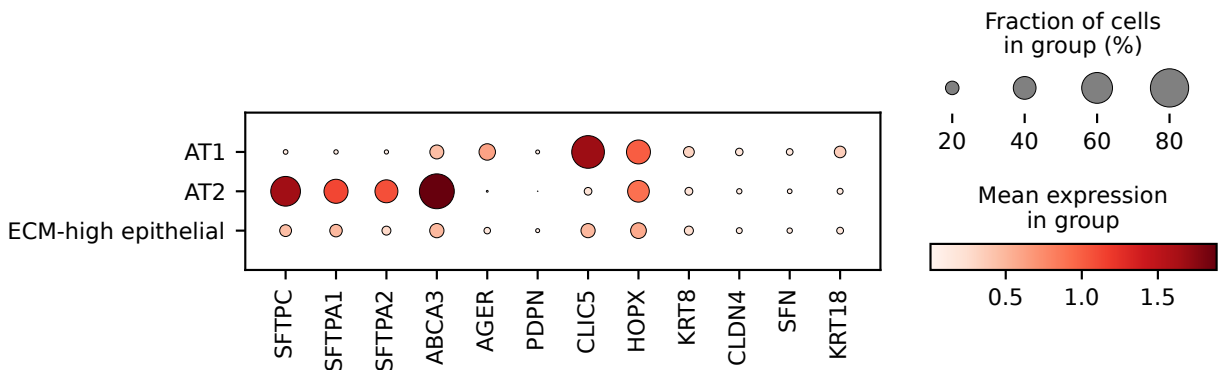


Figure 2: **Figure S2: Marker gene expression.** Expression of canonical AT1 (AGER, HOPX), AT2 (SFTPC, SFTPA1), and transitional (KRT8, CLDN4) markers across annotated cell types, confirming annotation quality.

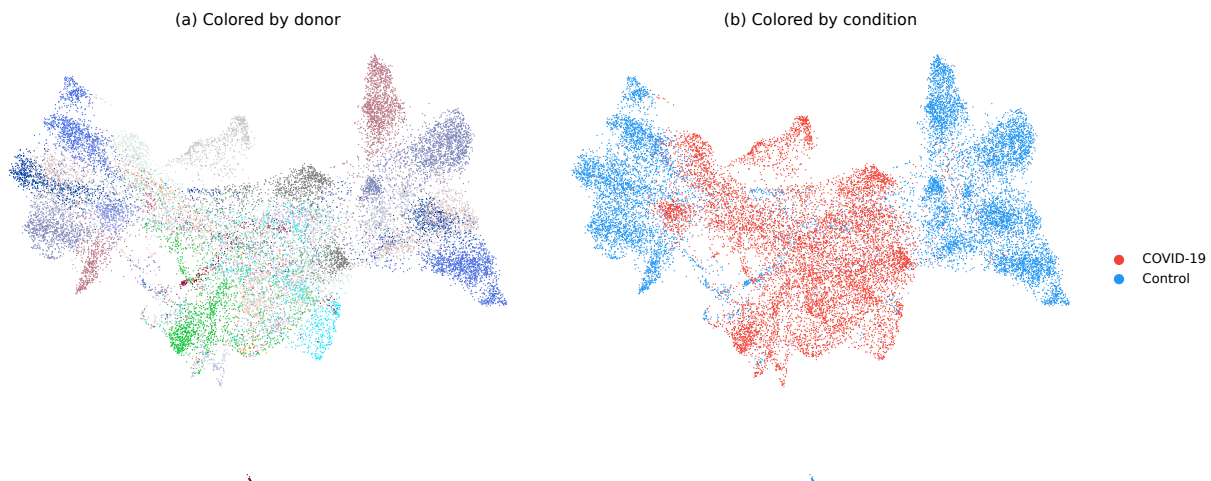


Figure 3: **Figure S3: Batch structure.** UMAP colored by donor ID and Harmony-corrected embedding, showing batch correction convergence and its effect on pseudotime estimation.

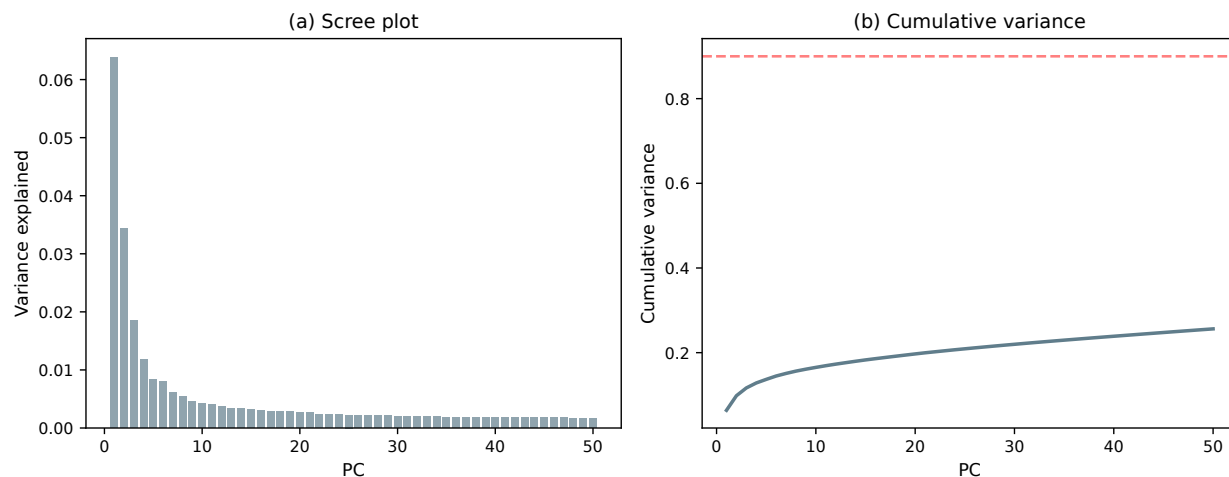


Figure 4: **Figure S4: PCA variance explained.** Scree plot and cumulative variance for the 50-component PCA used as input to the diffusion map and dispersion analyses.

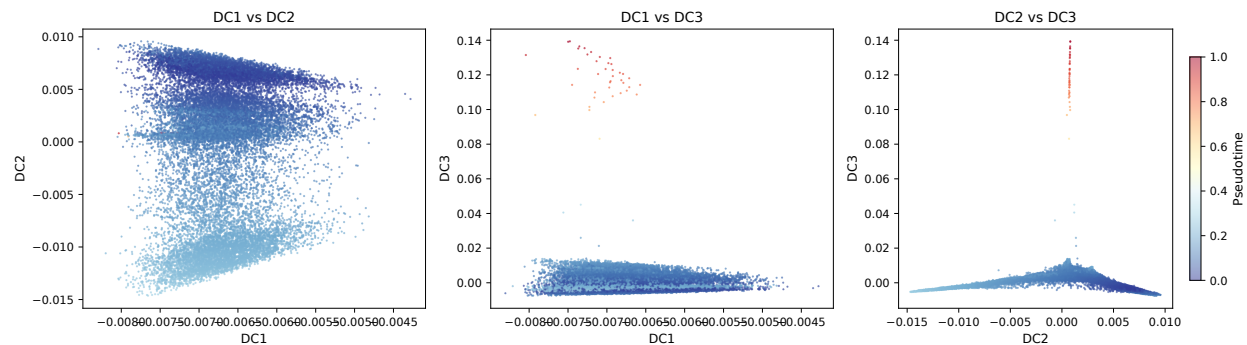


Figure 5: **Figure S5: Diffusion map components.** First three diffusion components colored by condition, cell type, and pseudotime, showing the manifold structure underlying the pseudotime ordering.

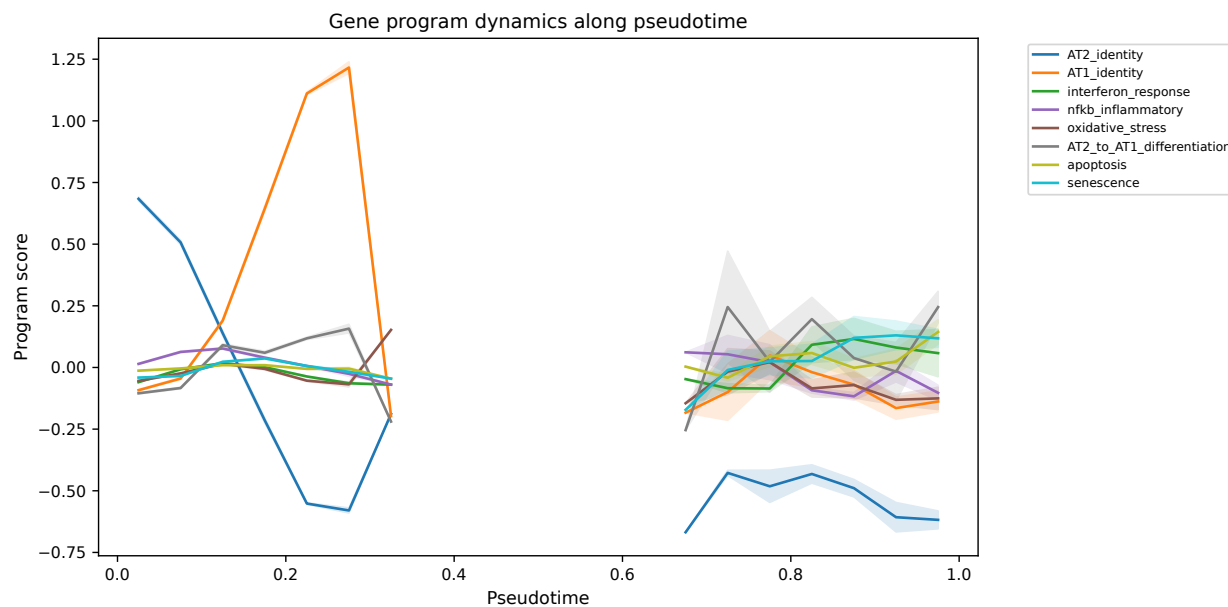


Figure 6: **Figure S6: Gene-program dynamics along pseudotime.** Mean program scores across 20 equal-width pseudotime bins for all eight gene programs, stratified by condition. Homeostatic programs (AT2, AT1 identity) decline with pseudotime; injury programs rise in the upper half.

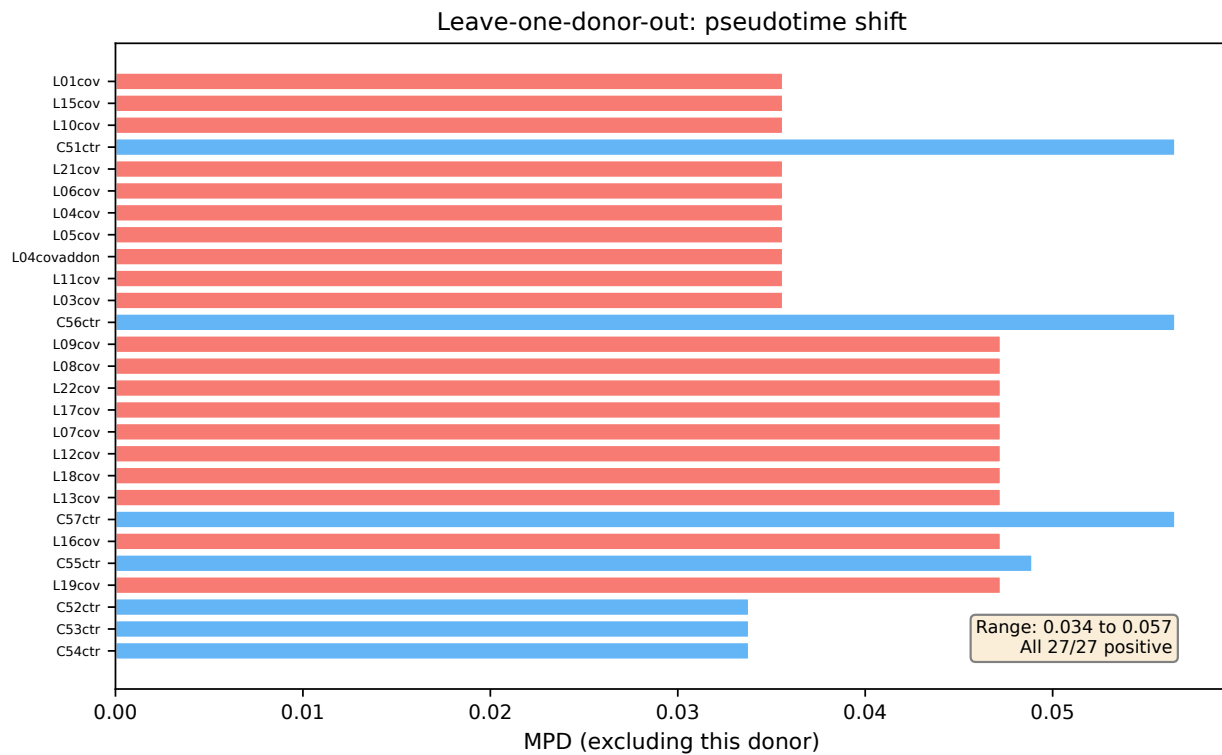


Figure 7: **Figure S9: Leave-one-donor-out analysis.** Median pseudotime difference for each of 27 iterations, confirming that no single donor drives the pseudotime shift (range 0.028–0.202; all positive).

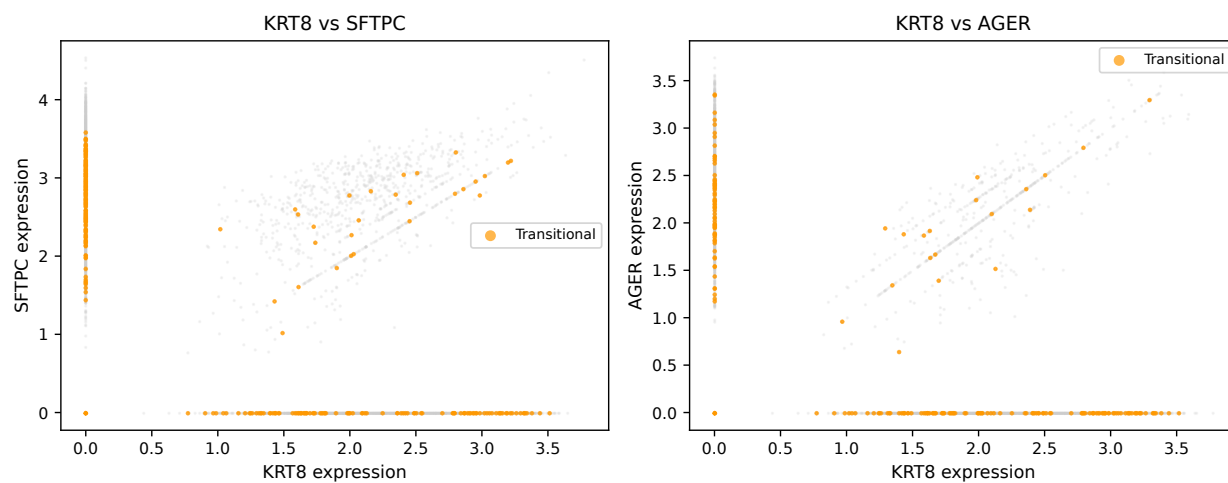


Figure 8: **Figure S10: Transitional cell characterization.** Extended marker analysis and pseudotime distribution for the ECM-high transitional compartment.

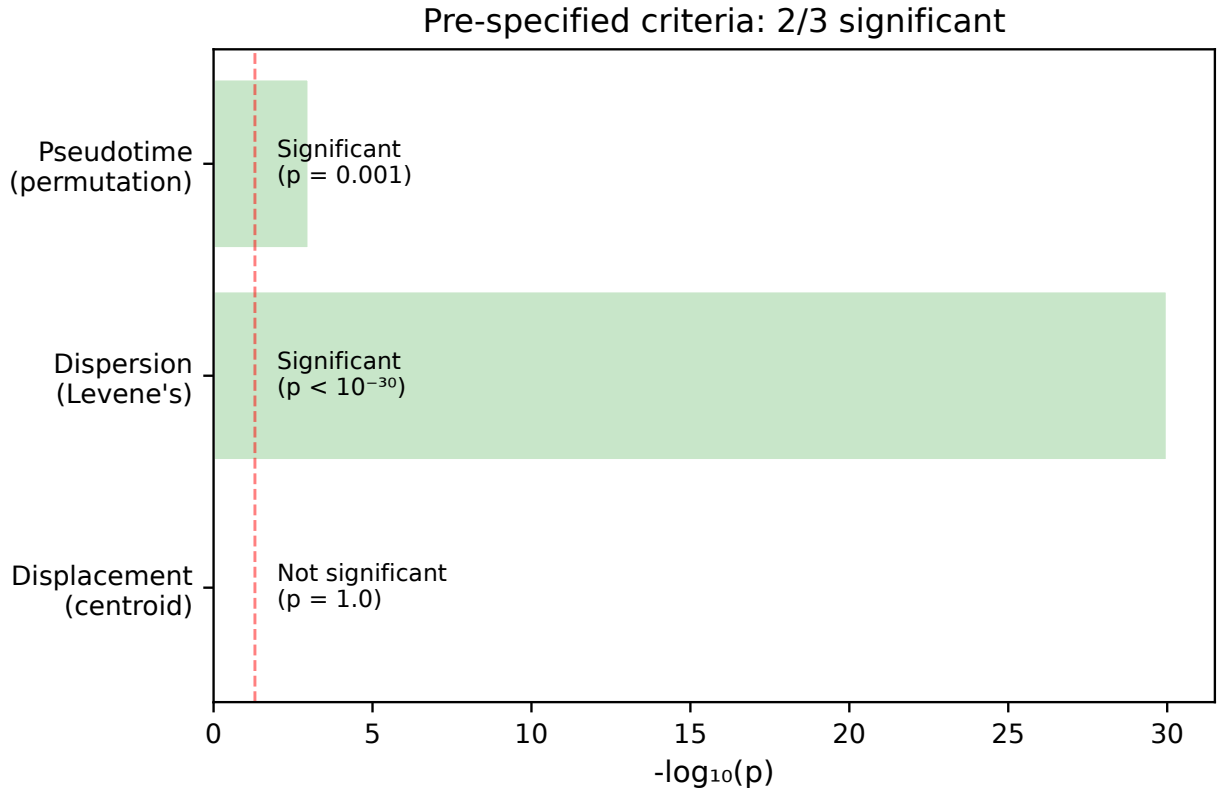


Figure 9: **Figure S11: Injury composite score.** Distribution of the six-program injury composite score by condition in both primary and replication cohorts.

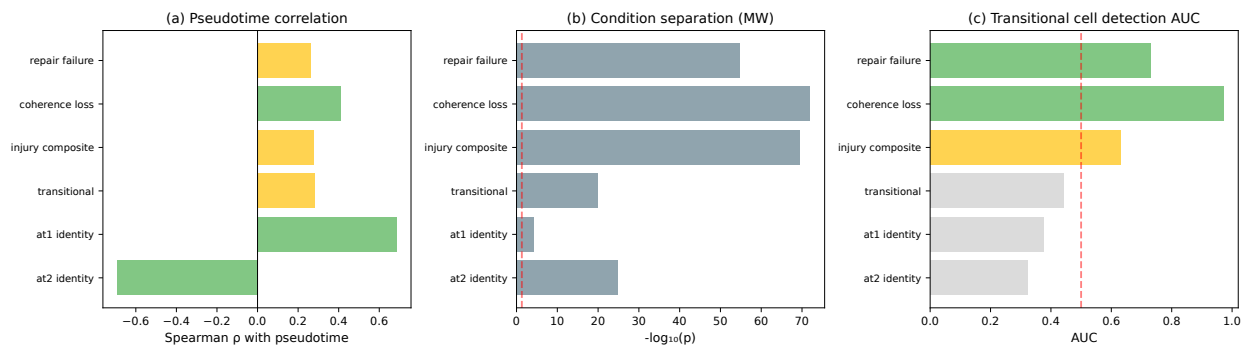


Figure 10: **Figure S12: Portable state scores.** Distributions of all six continuous state scores (AT2 identity, AT1 identity, transitional, injury composite, coherence loss, repair failure) by condition, with benchmark statistics.

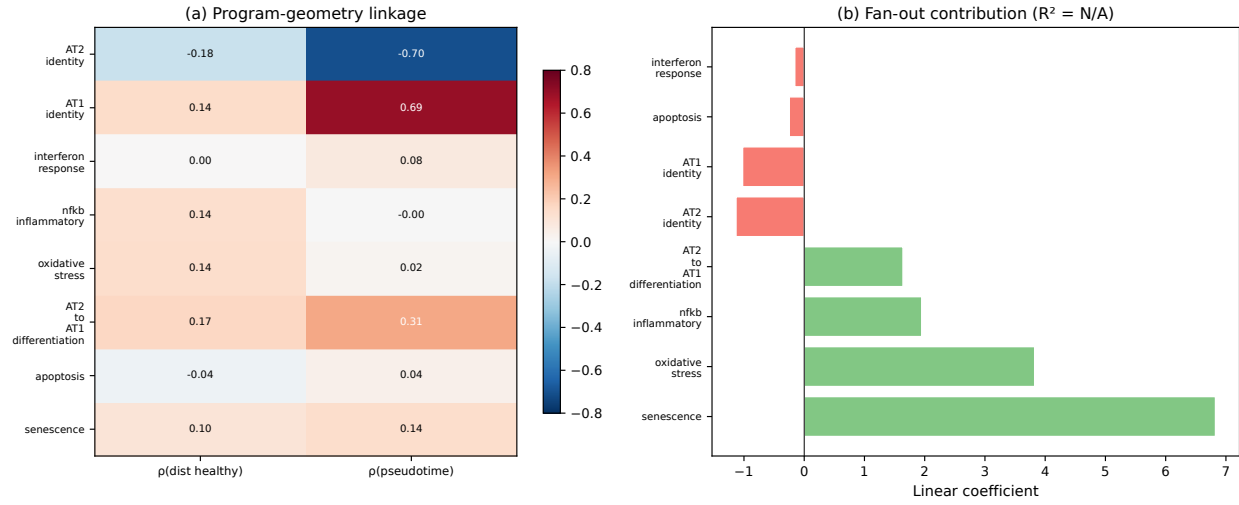


Figure 11: **Figure S13: Mechanistic decomposition.** (a) Program-geometry linkage: Spearman correlations of each program with distance-to-healthy-centroid and pseudotime. (b) Fan-out regression coefficients ($R^2 = 0.115$). (c) Repair-stall analysis: differentiation/apoptosis ratio across pseudotime bins by condition.